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M3 Q1. Apply the concepts of Ethereum and Bitcoin to identify a suitable platform for implementing a decentralized application. Justify your selection with relevant features.

Answer: For implementing a Decentralized Application (DApp), Ethereum is a more suitable platform than Bitcoin because it supports smart contracts and programmable blockchain applications.

Comparison between Bitcoin and Ethereum

Feature	**Bitcoin**	**Ethereum**
Primary Purpose	Digital cryptocurrency	Decentralized application platform
Smart Contracts	Limited scripting functionality	Supports programmable smart contracts
Consensus Mechanism	Proof of Work (PoW)	Proof of Stake (PoS)
Programming	Bitcoin Script	Solidity, Vyper
Flexibility	Limited programmability	Highly flexible and programmable
Applications	Digital payments, value transfer	DeFi, NFTs, DAOs, supply chain, DApps
Token Creation	Limited	Supports ERC-20, ERC-721, etc.

Why Ethereum is suitable for DApp development

1. Smart Contracts: Ethereum allows developers to write and deploy smart contracts that automatically execute predefined conditions without intermediaries.

2. Decentralization: Applications run on a distributed network of nodes, reducing dependence on centralized servers.

3. Solidity Support: Ethereum supports Solidity, a programming language used to develop smart contracts.

4. EVM (Ethereum Virtual Machine): EVM executes smart contracts consistently across the Ethereum network.

5. Token Standards: Ethereum supports standards such as ERC-20 for fungible tokens and ERC-721 for NFTs.

6. Security and Transparency: Transactions are recorded on an immutable public ledger and can be independently verified.

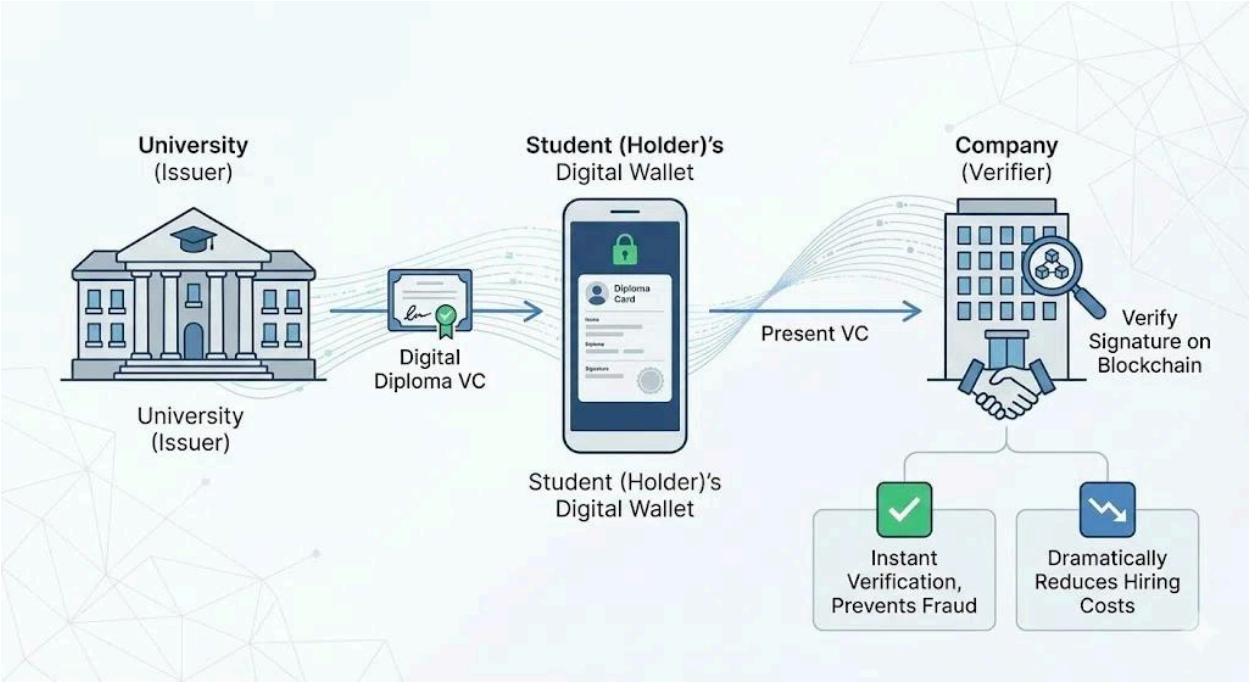
7. Large Ecosystem: Ethereum has extensive developer tools, libraries, wallets and frameworks for building DApps.

Ethereum is the preferred platform for developing decentralized applications because of its smart contract functionality, programmability, token standards and strong developer ecosystem. Bitcoin is primarily designed for peer-to-peer digital payments, whereas Ethereum provides a flexible environment for building complex decentralized applications.

M5 Q5. A university wants to issue blockchain-based digital degrees that employers can verify independently. Explain how blockchain can support certificate verification.

Answer: Blockchain technology can provide a secure, transparent and tamper-resistant system for issuing and verifying digital academic certificates. It eliminates the need for employers to contact universities manually for certificate verification.

Working of Blockchain-Based Certificate Verification



Step 1: Student Registration

- The university registers students and maintains their academic records.
- After successful completion of the course, the university prepares a digital degree certificate.

Step 2: Certificate Generation

- The university generates a digital certificate containing details such as:

- Student name
- Registration number
- Degree name
- Branch and specialization
- Graduation year
- University details

Step 3: Hash Generation

- A cryptographic hash (e.g., SHA-256) is generated from the certificate.
- This hash acts as a unique digital fingerprint of the certificate.
- The actual certificate can be stored securely off-chain, while its hash and verification details are recorded on the blockchain.

Step 4: Blockchain Registration

- The university uses its authorized blockchain account to register the certificate.
- A smart contract stores the certificate hash, student identifier, issuer details and issuance timestamp.
- Once recorded, the information becomes tamper-resistant.

Step 5: Certificate Sharing

- The student receives a digital certificate containing a QR code or unique certificate ID.
- The student shares it with potential employers.

Step 6: Employer Verification

- The employer scans the QR code or enters the certificate ID.
- The verification application retrieves the certificate record from the blockchain.
- It generates a hash of the submitted certificate and compares it with the stored hash.
- If both hashes match and the certificate has not been revoked, the certificate is considered authentic.

Role of Blockchain Components

Component	**Role**
Blockchain	Stores tamper-resistant certificate records
Smart Contract	Automates certificate issuance and verification
Cryptographic Hash	Detects modification of certificates
Digital Signature	Confirms that the certificate was issued by the authorized university
QR Code	Provides quick access to verification details
Distributed Ledger	Allows independent verification without contacting the university

Advantages

1. Tamper Resistance: Certificates cannot be secretly modified without detection.
2. Independent Verification: Employers can verify certificates without depending on manual university processes.
3. Fraud Prevention: Prevents fake degrees and forged academic documents.
4. Transparency: Certificate issuance records can be independently checked.
5. Time Efficiency: Reduces the time required for background verification.
6. Data Security: Sensitive student information can remain off-chain, with only hashes and necessary verification metadata stored on-chain.

Blockchain-based digital certificates provide a reliable and efficient verification mechanism. By combining smart contracts, cryptographic hashing and digital signatures, universities can issue tamper-resistant degrees that employers can independently authenticate.

M6 Q14. Design a Decentralized Autonomous Organization (DAO) for managing a community fund using blockchain-based governance rules.

Answer: A Decentralized Autonomous Organization (DAO) is an organization managed through blockchain-based smart contracts. Its rules and decisions are executed transparently without requiring a centralized authority.

A DAO can be designed to manage a community fund where members collectively decide how money should be allocated to development projects, social activities and community welfare.

Proposed DAO: Community Fund Management DAO

Objective: To create a decentralized platform where community members contribute funds, propose projects, vote on proposals and transparently monitor fund utilization.



1. Main Components

A. Community Members

- Individuals who join the DAO.
- Members can contribute funds and participate in governance.
- Eligible members can submit proposals and vote.

B. Smart Contracts

- Define governance rules.
- Manage fund deposits and withdrawals.
- Validate proposals and voting results.
- Automatically execute approved decisions.

C. Community Treasury

- A shared blockchain wallet that stores community contributions.
- Funds are released only when predefined governance conditions are satisfied.

D. Governance Tokens

- Members receive governance tokens based on predefined membership rules.
- Tokens provide voting rights in community decisions.

E. Voting Mechanism

- Members vote on proposals.
- Voting results are recorded transparently on the blockchain.

2. Working of the DAO

Step 1: Member Registration and Fund Contribution

- Community members register with the DAO.
- Members contribute cryptocurrency or supported digital tokens to the community treasury.
- Governance tokens are issued according to the DAO's membership rules.

Step 2: Proposal Creation

- Any eligible member can submit a proposal for using community funds.
- Example proposals:
    - Building a community library.

    - Organizing educational workshops.
    - Improving public infrastructure.
    - Providing financial assistance to needy members.

Step 3: Proposal Verification

- The smart contract checks whether the proposal meets predefined conditions.
- It verifies the member's eligibility and proposed funding amount.

Step 4: Community Voting

- Eligible members vote for or against the proposal.
- Voting takes place within a predefined period.
- Each member's voting power is determined by the governance rules.

Step 5: Decision Making

- After voting ends, the smart contract calculates the results.
- If the proposal receives the required minimum votes and approval percentage, it is accepted.
- Otherwise, the proposal is rejected.

Step 6: Fund Allocation

- The smart contract automatically releases the approved amount from the community treasury.
- Funds are transferred to the approved recipient or project wallet.

Step 7: Monitoring and Transparency

- All contributions, proposals, voting records and fund transfers are recorded on the blockchain.
- Community members can independently monitor how funds are being utilized.

3. Blockchain-Based Governance Rules

The DAO can implement the following rules through smart contracts:

Governance Rule	**Description**

Membership	Only registered members can participate in governance.
Proposal Submission	Eligible members can submit funding proposals.
Voting Period	Voting remains open for a fixed period, such as 7 days.
Voting Power	Each eligible member receives one vote to ensure equal participation.
Quorum	At least 30% of eligible members must participate for a valid decision.
Approval Rule	A proposal requires more than 50% approval of valid votes cast.
Fund Release	Funds are released only after successful proposal approval.
Transparency	All transactions and voting results are publicly verifiable.

Note: These are proposed governance parameters and can be modified according to community requirements.

4. Example Scenario

Suppose a community DAO has collected ₹10,00,000 in its treasury.

A member submits a proposal requesting ₹2,00,000 to build a community library.

- Total eligible members: 100
- Members participating in voting: 60
- Votes in favour: 45
- Votes against: 15
- Quorum required: 30 members
- Approval required: More than 50%

Since 60 members participated and 75% voted in favour, the proposal was **approved**.

The smart contract automatically releases ₹2,00,000 to the approved project wallet.

Example DAO fund allocation

5. Advantages of the DAO

1. Decentralized Decision Making: No single person controls the community fund.
2. Transparency: All financial transactions and voting records are publicly accessible.
3. Automation: Smart contracts automatically execute approved decisions.
4. Accountability: Members can track fund utilization.
5. Security: Blockchain records are tamper-resistant.
6. Fair Participation: Every eligible member can participate in governance.

A Community Fund Management DAO provides a transparent, secure and decentralized way to manage community finances. By using smart contracts, governance tokens and blockchain-based voting, it ensures collective decision-making, reduces the possibility of fund misuse and promotes accountability among community members.